λ_{SCE}: Separate Compilation, Capabilities, and First-Class Environments

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1 Core Language λ_E

1.1 Syntax

Types	A, B	$::=$	$\mathsf{Int} \mid \varepsilon \mid A \to B \mid A \,\&\, B \mid \{l : A\}$
Expressions	e, v	$::=$	$? \mid \varepsilon \mid i \mid \underline{n} \mid \lambda\{A\}.\, e \mid e_1\ e_2 \mid e_1\ ,,\ e_2$
		$\mid$	$e_1 \triangleright e_2 \mid \langle v,\, \lambda\, A.\, e\rangle \mid \{l = e\} \mid e.l \mid e.n$
Contexts	Γ	$::=$	$\varepsilon \mid \Gamma \,\&\, A$

1.2 Typing

$\boxed{\Gamma \vdash e : A}$ Typing

$$\frac{}{\Gamma \vdash ? : \Gamma}\ \textsc{Typ_ctx}$$

$$\frac{}{\Gamma \vdash i : \mathsf{Int}}\ \textsc{Typ_lit}$$

$$\frac{}{\Gamma \vdash \varepsilon : \varepsilon}\ \textsc{Typ_top}$$

$$\frac{}{\Gamma \vdash \underline{n} : \mathsf{lookup}(\Gamma, n)}\ \textsc{Typ_var}$$

$$\frac{\Gamma \vdash e_1 : A \qquad \Gamma \,\&\, A \vdash e_2 : B}{\Gamma \vdash e_1\ ,,\ e_2 : A \,\&\, B}\ \textsc{Typ_merge}$$

$$\frac{\Gamma \vdash e_1 : A \to B \qquad \Gamma \vdash e_2 : A}{\Gamma \vdash e_1\, e_2 : B}\ \textsc{Typ_app}$$

$$\frac{\Gamma \,\&\, A \vdash e : B}{\Gamma \vdash \lambda A.\, e : A \to B}\ \textsc{Typ_lam}$$

$$\frac{\Gamma \vdash e_1 : \Gamma_1 \qquad \Gamma_1 \vdash e_2 : A}{\Gamma \vdash e_1 \triangleright e_2 : A}\ \textsc{Typ_box}$$

$$\frac{\varepsilon \vdash v : \Gamma_1 \qquad \Gamma_1 \,\&\, A \vdash e : B}{\Gamma \vdash \langle v, \lambda A.\, e\rangle : A \to B}\ \textsc{Typ_clos}$$

$$\frac{\Gamma \vdash e : B \qquad \mathsf{lookup}(B, n) = A}{\Gamma \vdash e.n : A}\ \textsc{Typ_proj}$$

$$\frac{\Gamma \vdash e : A}{\Gamma \vdash \{l = e\} : \{l : A\}}\ \textsc{Typ_rcd}$$

$$\frac{\Gamma \vdash e : B \quad l : A \in B}{\Gamma \vdash e.l : A} \quad \textsc{Typ_sel}$$

1.3 Operational Semantics

Selection

$\boxed{v.l \rightsquigarrow v'}$ Selection

$$\frac{}{\{l = v\}.l \rightsquigarrow v} \quad \textsc{sel_rec}$$

$$\frac{v_1.l \rightsquigarrow v}{(v_1 \ ,, \ v_2).l \rightsquigarrow v} \quad \textsc{sel_mrgl}$$

$$\frac{v_2.l \rightsquigarrow v}{(v_1 \ ,, \ v_2).l \rightsquigarrow v} \quad \textsc{sel_mrgr}$$

Small-step

$\boxed{v \vdash e \hookrightarrow e'}$ Small-step Operational Semantics

$$\frac{v \vdash e \hookrightarrow e'}{v \vdash F[e] \hookrightarrow F[e']} \quad \textsc{Step_eval}$$

$$\frac{}{v \vdash ? \hookrightarrow v} \quad \textsc{Step_ctx}$$

$$\frac{}{v \vdash \underline{n} \hookrightarrow \mathsf{lookupv}(v, n)} \quad \textsc{Step_var}$$

$$\frac{v \ ,, \ v_1 \vdash e_2 \hookrightarrow e_2'}{v \vdash v_1 \ ,, \ e_2 \hookrightarrow v_1 \ ,, \ e_2'} \quad \textsc{Step_merger}$$

$$\frac{}{v \vdash \lambda A.\, e \hookrightarrow \langle v, \lambda A.\, e \rangle} \quad \textsc{Step_closure}$$

$$\frac{}{v \vdash \langle v_1, \lambda A.\, e \rangle\, v_2 \hookrightarrow (v_1 \ ,, \ v_2) \triangleright e} \quad \textsc{Step_beta}$$

$$\frac{v_1 \vdash e \hookrightarrow e'}{v \vdash v_1 \triangleright e \hookrightarrow v_1 \triangleright e'} \quad \textsc{Step_box}$$

$$\frac{}{v \vdash v_1 \triangleright v_2 \hookrightarrow v_2} \quad \textsc{Step_boxv}$$

$$\frac{}{v \vdash v_1.n \hookrightarrow \mathsf{lookupv}(v_1, n)} \quad \textsc{Step_proj}$$

$$\frac{v_1.l \rightsquigarrow v_2}{v \vdash v_1.l \hookrightarrow v_2} \quad \textsc{Step_sel}$$

Big-step

$\boxed{v \vdash e \Rightarrow e'}$ Big-step Operational Semantics

$$\frac{}{v \vdash ? \Rightarrow v} \quad \textsc{Bstep_ctx}$$

$$\frac{}{v \vdash \underline{n} \Rightarrow \mathsf{lookupv}(v, n)} \quad \textsc{Bstep_var}$$

$$\frac{}{v \vdash i \Rightarrow i} \quad \textsc{Bstep_lit}$$

$$\frac{}{v \vdash \varepsilon \Rightarrow \varepsilon} \quad \textsc{Bstep_unit}$$

$$\frac{}{v \vdash \lambda A.\, e \Rightarrow \langle v, \lambda A.\, e\rangle} \quad \textsc{Bstep_lam}$$

$$\frac{}{v \vdash \langle v_1, \lambda A.\, e\rangle \Rightarrow \langle v_1, \lambda A.\, e\rangle} \quad \textsc{Bstep_clos}$$

$$\frac{\begin{array}{c} v \vdash e_1 \Rightarrow v_1 \\ v_1 \vdash e_2 \Rightarrow v_2 \end{array}}{v \vdash e_1 \triangleright e_2 \Rightarrow v_2} \quad \textsc{Bstep_box}$$

$$\frac{v \vdash e \Rightarrow v_1}{v \vdash e.n \Rightarrow \mathsf{lookupv}(v_1, n)} \quad \textsc{Bstep_proj}$$

$$\frac{\begin{array}{c} v \vdash e_1 \Rightarrow v_1 \\ v \;,,\; v_1 \vdash e_2 \Rightarrow v_2 \end{array}}{v \vdash e_1 \;,,\; e_2 \Rightarrow v_1 \;,,\; v_2} \quad \textsc{Bstep_merge}$$

$$\frac{\begin{array}{c} v \vdash e_1 \Rightarrow \langle v_1, \lambda A.\, e\rangle \\ v \vdash e_2 \Rightarrow v_2 \\ v_1 \;,,\; v_2 \vdash e \Rightarrow v' \end{array}}{v \vdash e_1\, e_2 \Rightarrow v'} \quad \textsc{Bstep_app}$$

$$\frac{v \vdash e \Rightarrow v_1}{v \vdash \{l = e\} \Rightarrow \{l = v_1\}} \quad \textsc{Bstep_rec}$$

$$\frac{\begin{array}{c} v \vdash e \Rightarrow v_1 \\ v_1.l \rightsquigarrow v_2 \end{array}}{v \vdash e.l \Rightarrow v_2} \quad \textsc{Bstep_sel}$$

2 Source Language λ_{SCE}

2.1 Syntax

Module Types	$\mathcal{M}$	::=	$\mathsf{Intf}\ \mathcal{A} \mid \mathcal{A} \to_m \mathcal{M}$
Source Types	$\mathcal{A}, \mathcal{B}$	::=	$\mathsf{Int} \mid \varepsilon \mid \mathcal{A} \to \mathcal{B} \mid \mathcal{A} \,\&\, \mathcal{B} \mid \{l : \mathcal{A}\} \mid \mathsf{Mod}\ \mathcal{M}$
Sandbox	σ	::=	$\mathsf{S} \mid \mathsf{O}$
Source Exp.	$\hat{e}, \hat{v}$	::=	$\hat{?} \mid \hat{\varepsilon} \mid \hat{i} \mid \hat{\lambda}\, \mathcal{A}.\, \hat{e} \mid \hat{e}_1\ \hat{e}_2 \mid \hat{e}_1\ \hat{,,}\ \hat{e}_2 \mid \hat{e}_1\ \hat{,,}_n\ \hat{e}_2 \mid \langle \hat{v},\ \hat{\lambda}\, \mathcal{A}.\, \hat{e}\rangle \mid \{l = \hat{e}\} \mid \hat{e}.l \mid \hat{e}.n \mid \mathsf{let}\ \hat{e}_1 : \mathcal{A}\ \mathsf{in}\ \hat{e}_2 \mid \mathsf{ope}$
		|	$\hat{e}_1$

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2.2 Source Values

$$\hat{v} \ ::= \ \hat{i} \mid \hat{\varepsilon} \mid \langle \hat{v},\ \hat{\lambda}\, \mathcal{A}.\, \hat{e}\rangle \mid \langle \hat{v},\ \mathcal{A},\ \hat{e}\rangle_m \mid \hat{v}_1\ \hat{,,}\ \hat{v}_2 \mid \hat{v}_1\ \hat{,,}_n\ \hat{v}_2 \mid \{l = \hat{v}\}$$

2.3 Type Elaboration

$[\![\cdot]\!]$ maps source types to core types:

$$\begin{array}{rcl} [\![\mathsf{Int}]\!] & = & \mathsf{Int} \\ [\![\varepsilon]\!] & = & \varepsilon \\ [\![\mathcal{A} \to \mathcal{B}]\!] & = & [\![\mathcal{A}]\!] \to [\![\mathcal{B}]\!] \\ [\![\mathcal{A} \,\&\, \mathcal{B}]\!] & = & [\![\mathcal{A}]\!] \,\&\, [\![\mathcal{B}]\!] \end{array} \qquad \begin{array}{rcl} [\![\{l : \mathcal{A}\}]\!] & = & \{l : [\![\mathcal{A}]\!]\} \\ [\![\mathsf{Mod}\ (\mathsf{Intf}\ \mathcal{A})]\!] & = & [\![\mathcal{A}]\!] \\ [\![\mathsf{Mod}\ (\mathcal{A} \to_m \mathcal{M})]\!] & = & [\![\mathcal{A}]\!] \to [\![\mathsf{Mod}\ \mathcal{M}]\!] \end{array}$$

2.4 Type-Directed Elaboration

Judgment: $\boxed{\mathcal{G} \vdash \hat{e} : \mathcal{A} \rightsquigarrow e}$

Core Expression Rules

$$\frac{}{\mathcal{G} \vdash \hat{?} : \mathcal{G} \rightsquigarrow ?}\ \text{E-Ctx} \qquad \frac{}{\mathcal{G} \vdash \hat{i} : \mathsf{Int} \rightsquigarrow i}\ \text{E-Lit} \qquad \frac{}{\mathcal{G} \vdash \hat{\varepsilon} : \varepsilon \rightsquigarrow \varepsilon}\ \text{E-Unit} \qquad \frac{\mathcal{G} \vdash \hat{e} : \mathcal{B} \rightsquigarrow e \qquad \mathsf{lookup}(\mathcal{B}, n) = \mathcal{A}}{\mathcal{G} \vdash \hat{e}.n : \mathcal{A} \rightsquigarrow e.n}\ \text{E-Proj}$$

$$\frac{\mathcal{G} \,\&\, \mathcal{A} \vdash \hat{e} : \mathcal{B} \rightsquigarrow e}{\mathcal{G} \vdash \hat{\lambda}\, \mathcal{A}.\, \hat{e} : \mathcal{A} \to \mathcal{B} \rightsquigarrow \lambda[\![\mathcal{A}]\!].\, e}\ \text{E-Lam} \qquad {}_2 : \mathcal{A} \rightsquigarrow e_1 \triangleright e_2$$

Module Rules

Helper for functor result types:

$$\mathsf{mkMT}(\mathcal{A}, \mathcal{B}) = \begin{cases} \mathcal{A} \to_m \mathcal{M} & \text{if } \mathcal{B} = \mathsf{Mod}\ \mathcal{M} \\ \mathcal{A} \to_m \mathsf{Intf}\ \mathcal{B} & \text{otherwise} \end{cases}$$

$$\frac{\varepsilon \vdash \hat{e} : \mathcal{B} \rightsquigarrow e}{\mathcal{G} \vdash \mathsf{struct}^{\mathsf{S}}\ \hat{e} : \mathsf{Mod}\ (\mathsf{Intf}\ \mathcal{B}) \rightsquigarrow \varepsilon \triangleright e}\ \text{E-Struct-S} \qquad \frac{\mathcal{G} \vdash \hat{e} : \mathcal{B} \rightsquigarrow e}{\mathcal{G} \vdash \mathsf{struct}^{\mathsf{O}}\ \hat{e} : \mathsf{Mod}\ (\mathsf{Intf}\ \mathcal{B}) \rightsquigarrow ? \triangleright e}\ \text{E-Struct-O}$$

$$\frac{\varepsilon \,\&\, \mathcal{A} \vdash \hat{e} : \mathcal{B} \rightsquigarrow e \qquad \mathcal{M} = \mathsf{mkMT}(\mathcal{A}, \mathcal{B})}{\mathcal{G} \vdash \mathsf{functor}^{\mathsf{S}}(\mathcal{A})\ \hat{e} : \mathsf{Mod}\ \mathcal{M} \rightsquigarrow \varepsilon \triangleright \lambda[\![\mathcal{A}]\!].\, e}\ \text{E-Functor-S} \qquad \frac{\mathcal{G} \,\&\, \mathcal{A} \vdash \hat{e} : \mathcal{B} \rightsquigarrow e \qquad \mathcal{M} = \mathsf{mkMT}(\mathcal{A}, \mathcal{B})}{\mathcal{G} \vdash \mathsf{functor}^{\mathsf{O}}(\mathcal{A})\ \hat{e} : \mathsf{Mod}\ \mathcal{M} \rightsquigarrow \lambda[\![\mathcal{A}]\!].\, e}\ \text{E-Functor-O}$$

$$\frac{\mathcal{G} \vdash \hat{e}_1 : \mathcal{G}' \rightsquigarrow e_1 \qquad \mathcal{G}' \,\&\, \mathcal{A} \vdash \hat{e}_2 : \mathcal{B} \rightsquigarrow e_2 \qquad \mathcal{M} = \mathsf{mkMT}(\mathcal{A}, \mathcal{B})}{\mathcal{G} \vdash \langle \hat{e}_1,\, \mathcal{A},\, \hat{e}_2 \rangle_m : \mathsf{Mod}\ \mathcal{M} \rightsquigarrow \langle e_1,\, \lambda\, [\![\mathcal{A}]\!].\, e_2 \rangle}\ \text{E-MClos}$$

$$\frac{\mathcal{G} \vdash \hat{e}_1 : \mathsf{Mod}\ (\mathcal{A} \to_m \mathcal{M}) \rightsquigarrow e_1 \qquad \mathcal{G} \vdash \hat{e}_2 : \mathcal{A} \rightsquigarrow e_2}{\mathcal{G} \vdash \mathsf{Link}(\hat{e}_1,\, \hat{e}_2) : \mathsf{Mod}\ \mathcal{M} \rightsquigarrow e_1\ e_2}\ \text{E-Link}$$

2.5 Source Selection

Judgment: $\boxed{\hat{v}.l \rightsquigarrow \hat{v}'}$

$$\frac{}{\{l = \hat{v}\}.l \rightsquigarrow \hat{v}}\ \text{SS-Rcd} \qquad \frac{\hat{v}_1.l \rightsquigarrow \hat{v}}{(\hat{v}_1\ \hat{,}\ \hat{v}_2).l \rightsquigarrow \hat{v}}\ \text{SS-DMrg-L} \qquad \frac{\hat{v}_2.l \rightsquigarrow \hat{v}}{(\hat{v}_1\ \hat{,}\ \hat{v}_2).l \rightsquigarrow \hat{v}}\ \text{SS-DMrg-R} \qquad \frac{\hat{v}_1.l \rightsquigarrow \hat{v}}{(\hat{v}_1\ \hat{,}_n\ \hat{v}_2).l \rightsquigarrow \hat{v}}\ \text{SS-NMrg-L} \qquad \frac{\hat{v}_2.l \rightsquigarrow \hat{v}}{(\hat{v}_1\ \hat{,}_n\ \hat{v}_2).l \rightsquigarrow \hat{v}}\ \text{SS-NMrg-R}$$

2.6 Source Big-Step Semantics

Judgment: $\boxed{\hat{v} \vdash \hat{e} \Rightarrow \hat{v}'}$

Base Cases

$$\frac{\mathsf{svalue}\ \hat{v}}{\hat{v} \vdash \hat{?} \Rightarrow \hat{v}}\ \text{S-Ctx} \qquad \frac{\mathsf{svalue}\ \hat{v}}{\hat{v} \vdash \hat{i} \Rightarrow \hat{i}}\ \text{S-Lit} \qquad \frac{\mathsf{svalue}\ \hat{v}}{\hat{v} \vdash \hat{\varepsilon} \Rightarrow \hat{\varepsilon}}\ \text{S-Unit} \qquad \frac{\mathsf{svalue}\ \hat{v} \qquad \mathsf{svalue}\ \hat{v}_1}{\hat{v} \vdash \langle \hat{v}_1,\, \hat{\lambda}\, \mathcal{A}.\, \hat{e} \rangle \Rightarrow \langle \hat{v}_1,\, \hat{\lambda}\, \mathcal{A}.\, \hat{e} \rangle}\ \text{S-ClosV}$$

$$\frac{\mathsf{svalue}\ \hat{v} \qquad \mathsf{svalue}\ \hat{v}_1}{\hat{v} \vdash \langle \hat{v}_1,\, \mathcal{A},\, \hat{e} \rangle_m \Rightarrow \langle \hat{v}_1,\, \mathcal{A},\, \hat{e} \rangle_m}\ \text{S-MClosV}$$

Expression Rules

$$\frac{\mathsf{svalue}\ \hat{v}}{\hat{v} \vdash \hat{\lambda}\,\mathcal{A}.\,\hat{e} \Rightarrow \langle \hat{v},\ \hat{\lambda}\,\mathcal{A}.\,\hat{e}\rangle}\ \text{S-Lam} \qquad \frac{\hat{v} \vdash \hat{e} \Rightarrow \hat{v}' \qquad \mathsf{lookupv}(\hat{v}', n) = \hat{v}''}{\hat{v} \vdash \hat{e}.n \Rightarrow \hat{v}''}\ \text{S-Proj} \qquad {}_2 \Rightarrow \hat{v}'$$

Module Rules

$$\frac{\mathsf{svalue}\ \hat{v} \qquad \hat{\varepsilon} \vdash \hat{e} \Rightarrow \hat{v}'}{\hat{v} \vdash \mathsf{struct}^{\mathsf{S}}\ \hat{e} \Rightarrow \hat{v}'}\ \text{S-Struct-S} \qquad \frac{\hat{v} \vdash \hat{e} \Rightarrow \hat{v}'}{\hat{v} \vdash \mathsf{struct}^{\mathsf{O}}\ \hat{e} \Rightarrow \hat{v}'}\ \text{S-Struct-O} \qquad \frac{\mathsf{svalue}\ \hat{v}}{\hat{v} \vdash \mathsf{functor}^{\mathsf{S}}(\mathcal{A})\ \hat{e} \Rightarrow \langle \hat{\varepsilon},\ \mathcal{A},\ \hat{e}\rangle_m}\ \text{S-Functor-S}$$

$$\frac{\mathsf{svalue}\ \hat{v}}{\hat{v} \vdash \mathsf{functor}^{\mathsf{O}}(\mathcal{A})\ \hat{e} \Rightarrow \langle \hat{v},\ \mathcal{A},\ \hat{e}\rangle_m}\ \text{S-Functor-O} \qquad \frac{\hat{v} \vdash \hat{e}_1 \Rightarrow \langle \hat{v}_1,\ \mathcal{A},\ \hat{e}\rangle_m \qquad \hat{v} \vdash \hat{e}_2 \Rightarrow \hat{v}_2 \qquad \hat{v}_1\ \hat{,}\ \hat{v}_2 \vdash \hat{e} \Rightarrow \hat{v}'}{\hat{v} \vdash \mathsf{Link}(\hat{e}_1,\ \hat{e}_2) \Rightarrow \hat{v}'}\ \text{S-Link}$$

$$\frac{\hat{v} \vdash \hat{e}_1 \Rightarrow \langle \hat{v}_1,\ \hat{\lambda}\,\mathcal{A}.\,\hat{e}\rangle \qquad \hat{v} \vdash \hat{e}_2 \Rightarrow \hat{v}_2 \qquad \hat{v}_1\ \hat{,}\ \hat{v}_2 \vdash \hat{e} \Rightarrow \hat{v}'}{\hat{v} \vdash \mathsf{Link}(\hat{e}_1,\ \hat{e}_2) \Rightarrow \hat{v}'}\ \text{S-Link-C}$$

3 Auxiliary Judgments

3.1 Conflict

$\boxed{A \oslash B}$ Conflict

$$\frac{}{\{l : A\} \oslash \{l : B\}}\ \text{CFL_RCD}$$

$$\frac{A \oslash C}{A \,\&\, B \oslash C}\ \text{CFL_ANDL}$$

$$\frac{B \oslash C}{A \,\&\, B \oslash C}\ \text{CFL_ANDR}$$

$$\frac{A \oslash B}{A \oslash B \,\&\, C}\ \text{CFL_RANDL}$$

$$\frac{A \oslash C}{A \oslash B \,\&\, C}\ \text{CFL_RANDR}$$

3.2 Uniqueness

$\boxed{\mathsf{ok}\ A}$ Uniqueness

$$\frac{}{\mathsf{ok}\ \{l : A\}}\ \text{UQ_RCD}$$

$$\frac{\begin{array}{l}\mathsf{ok}\ A\\ \mathsf{ok}\ B\\ \neg(A \oslash B)\end{array}}{\mathsf{ok}\ A \,\&\, B}\ \text{UQ_AND}$$

3.3 Core Containment

$\boxed{l : A \in B}$ Containment

$$\frac{}{l : A \in \{l : A\}}\ \text{CTM_RCD}$$

$$\frac{\begin{array}{l} l : A \in B \\ l \notin \mathit{label}(C) \end{array}}{l : A \in B \,\&\, C}\ \text{CTM_ANDL}$$

$$\frac{\begin{array}{l} l : A \in C \\ l \notin \mathit{label}(B) \end{array}}{l : A \in B \,\&\, C}\ \text{CTM_ANDR}$$

4 Metatheory

All theorems are mechanized in Coq and validated via property-based testing in Haskell (500 tests each, all passing).

Theorem 4.1 (Type-Preserving Elaboration)**.** *If* $\mathcal{G} \vdash \hat{e} : \mathcal{A} \rightsquigarrow e$*, then* $[\![\mathcal{G}]\!] \vdash e : [\![\mathcal{A}]\!]$*.*

Theorem 4.2 (Uniqueness of Elaboration)**.** *If* $\mathcal{G} \vdash \hat{e} : \mathcal{A}_1 \rightsquigarrow e_1$ *and* $\mathcal{G} \vdash \hat{e} : \mathcal{A}_2 \rightsquigarrow e_2$*, then* $\mathcal{A}_1 = \mathcal{A}_2$ *and* $e_1 = e_2$*.*

Theorem 4.3 (Semantics Preservation)**.** *If* $\mathcal{G} \vdash \hat{e} : \mathcal{A} \rightsquigarrow e$*,* $\mathsf{svalue}\ \hat{\rho}$*,* $\hat{\rho} \vdash \hat{e} \Rightarrow \hat{v}$*, and* $\hat{\rho} \sim \rho$ *at type* $\mathcal{G}$*, then there exists* v *such that* $\rho \vdash e \Rightarrow v$ *and* $\hat{v} \sim v$ *at type* $\mathcal{A}$*.*

Theorem 4.4 (Separate Compilation)**.** *If* $\mathcal{G} \vdash \hat{e}_1 : \mathsf{Mod}\ (\mathcal{A} \rightarrow_m \mathcal{M}) \rightsquigarrow e_1$ *and* $\mathcal{G} \vdash \hat{e}_2 : \mathcal{A} \rightsquigarrow e_2$*, then* $\mathcal{G} \vdash \mathsf{Link}(\hat{e}_1, \hat{e}_2) : \mathsf{Mod}\ \mathcal{M} \rightsquigarrow e_1\ e_2$*.*

Theorem 4.5 (Sandbox Struct Isolation)**.** *If* $\mathsf{svalue}\ \hat{\rho}_1$*,* $\mathsf{svalue}\ \hat{\rho}_2$*,* $\hat{\rho}_1 \vdash \mathsf{struct}^{\mathsf{S}}\ \hat{e} \Rightarrow \hat{v}_1$*, and* $\hat{\rho}_2 \vdash \mathsf{struct}^{\mathsf{S}}\ \hat{e} \Rightarrow \hat{v}_2$*, then* $\hat{v}_1 = \hat{v}_2$*.*

Theorem 4.6 (Sandbox Functor Isolation)**.** *If* $\mathsf{svalue}\ \hat{\rho}_1$ *and* $\mathsf{svalue}\ \hat{\rho}_2$*, then* $\hat{\rho}_1 \vdash \mathsf{functor}^{\mathsf{S}}(\mathcal{A})\ \hat{e} \Rightarrow \langle \hat{\varepsilon}, \mathcal{A}, \hat{e} \rangle_m$ *and* $\hat{\rho}_2 \vdash \mathsf{functor}^{\mathsf{S}}(\mathcal{A})\ \hat{e} \Rightarrow \langle \hat{\varepsilon}, \mathcal{A}, \hat{e} \rangle_m$*.*

Theorem 4.7 (Secure Linking)**.** *If a sandboxed functor's body is closed (does not depend on the argument), then the result of* Link *is independent of both the linking environment and the argument value.*

Theorem 4.8 (NMrg Independence)**.** *If* $\mathcal{G} \vdash \hat{e}_1\ \hat{,}_n\ \hat{e}_2 : \mathcal{A} \,\&\, \mathcal{B} \rightsquigarrow e$ *and* $\mathcal{G} \vdash \hat{e}_2 : \mathcal{B}' \rightsquigarrow e_2$*, then* $\mathcal{B} = \mathcal{B}'$*.*

Theorem 4.9 (Full Pipeline)**.** *If* $\varepsilon \vdash \hat{e} : \mathcal{A} \rightsquigarrow e$*, then* $[\![\varepsilon]\!] \vdash e : [\![\mathcal{A}]\!]$ *and there exists a value* v *such that* $\varepsilon \vdash e \hookrightarrow^* v$*.*